

Standard QM (Griffiths, Sakurai)

$$\mathcal{H} = \mathcal{H}_R \otimes \mathcal{H}_L$$

$$|\Psi\rangle = \sum_n c_n |n\rangle_R |n\rangle_L$$

$$\rho_R = \text{Tr}_L |\Psi\rangle\langle\Psi|$$

$$S_R = -\text{Tr} \rho_R \log \rho_R$$

subsystem $\equiv$ tensor factor

generalizes

Algebraic QM (von Neumann, Liu)

$\mathcal{M} \subset B(\mathcal{H})$, a weakly closed $*$ -algebra

$$\mathcal{M}' = \{B : [B, A] = 0 \ \forall A \in \mathcal{M}\}$$

type I/II: $\rho_{\mathcal{M}}$ via $\text{tr}(A\rho_{\mathcal{M}}) = \langle\Psi|A|\Psi\rangle$

type III: modular operator $\Delta_{\Psi} = e^{-K_{\Psi}}$

subsystem $\equiv$ operator subalgebra

special case: $\mathcal{M}$ is type I
 $\iff \mathcal{H} = \mathcal{H}_R \otimes \mathcal{H}_L$ exists